

Reliable wallet APIs

NovaWallet Ledger Service | Backend / API Developer (.NET) case study

**Move money once.
Keep the evidence.**

CORRECTNESS

Atomic balances and limits

RELIABILITY

Durable idempotent
outcomes

ACCOUNTABILITY

Separate append-only audit

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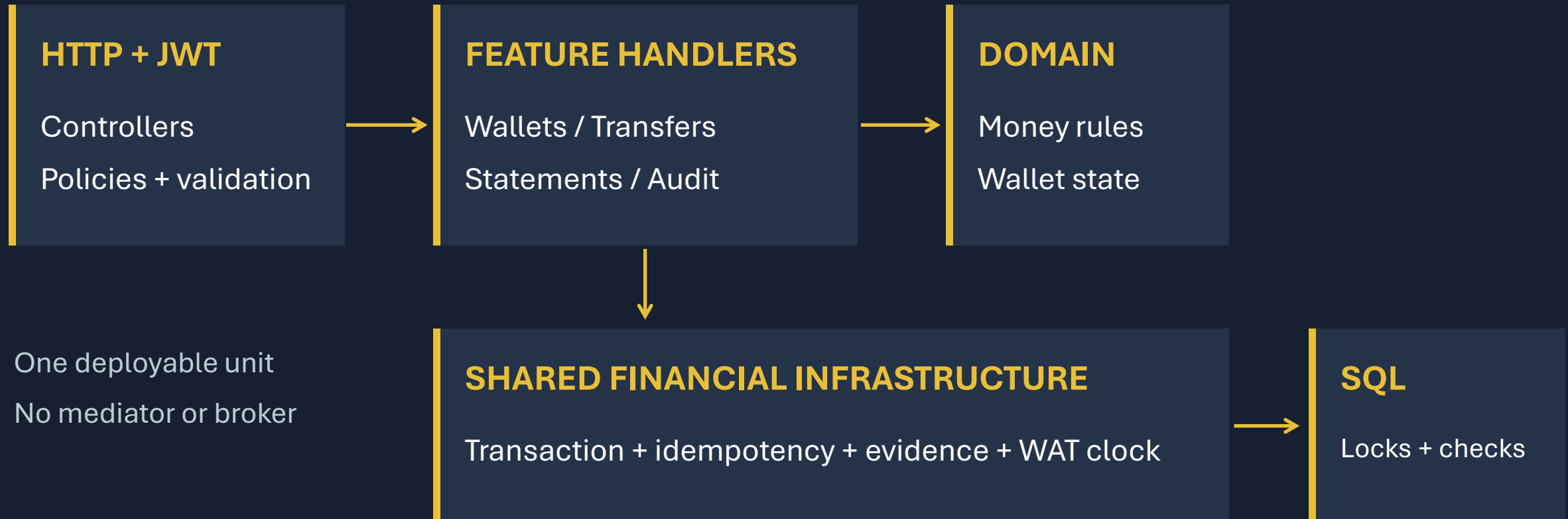
The brief becomes five invariants

Business rules drive the design, not the framework.

- | | | |
|-----------|-------------------------|---|
| 01 | Exact money | NGN only; long / bigint kobo; zero opening balance |
| 02 | Safe spending | Atomic debit + credit; no negative balances |
| 03 | Safe retries | Same key + payload = stored response; changed payload = 409 |
| 04 | Daily control | Outbound cap: NGN 500,000; reset at midnight WAT |
| 05 | Durable evidence | Financial history and audit commit with the movement |

One application. Clear feature boundaries.

Controller-based Vertical Slice Architecture | .NET 9 + EF Core 9 + SQL Server 2022



A small model with strong database rules

Operational balance + immutable movement evidence; not a full general ledger.

Wallet

CustomerId UNIQUE
BalanceKobo bigint ≥ 0
Currency = NGN

Transfer

Source + destination
AmountKobo > 0
CompletedAtUtc

IdempotencyRecord

Scope + key UNIQUE
Request hash
Stored HTTP outcome

One successful transfer: 1 transfer record + 2 history entries + 2 audit entries

WalletTransaction

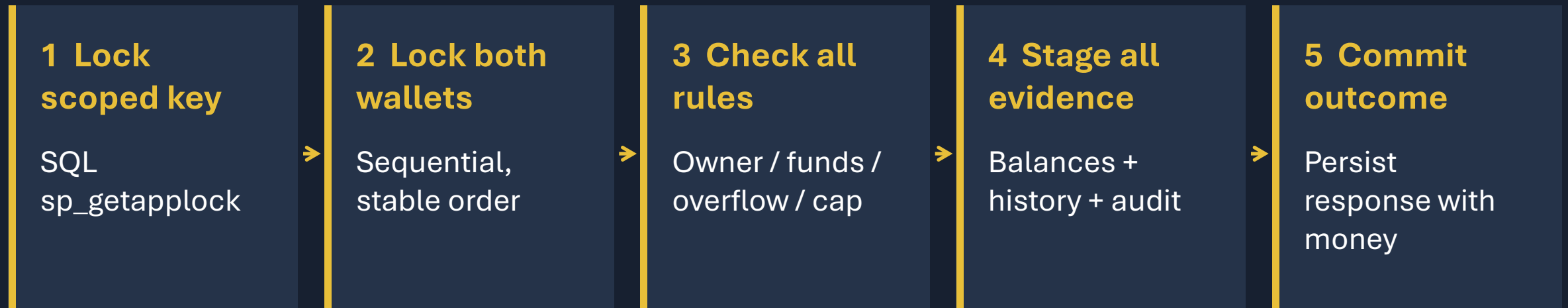
Operation + before/after
Newest-first statement

AuditEntry

Actor + trace + operation
Append-only evidence

The transfer is one indivisible SQL transaction

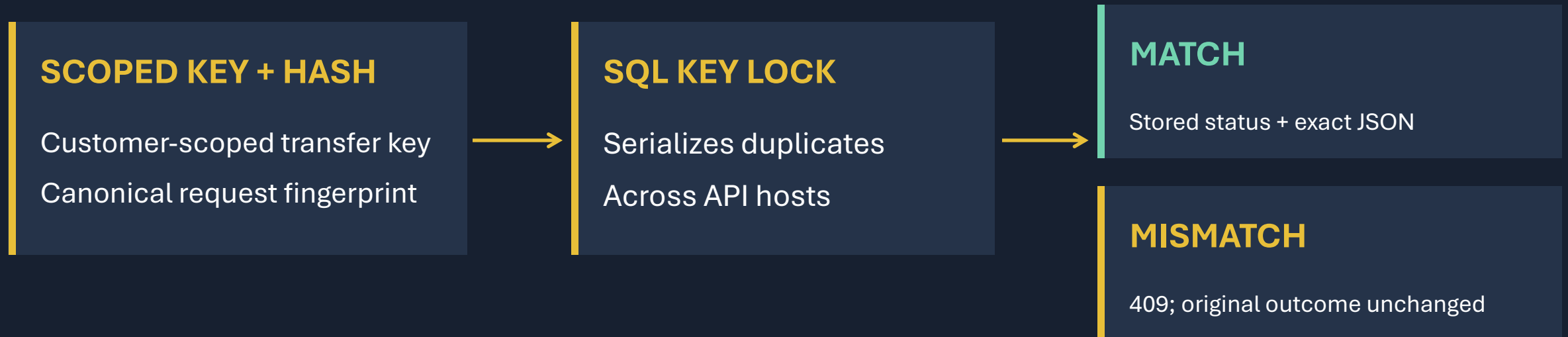
READ COMMITTED + explicit transaction-owned locks; never a process-local mutex.



ALL COMMIT TOGETHER / ALL ROLL BACK TOGETHER

Idempotency survives HTTP retries and API replicas

The database stores the outcome; the key is not an in-memory cache.



A key identifies one intent, including its financial rejection.

Retry the same intent with the same key. Use a new key for a deliberate new attempt.

Daily limits and audit are inside the safety boundary

Business controls are enforced by the backend, not just the user interface.

WAT OUTBOUND CAP

NGN 500,000 = 50,000,000 kobo

Midnight WAT = 23:00 UTC

Window: start inclusive, end exclusive

Check completed outbound while locked

APPEND-ONLY EVIDENCE

Audit inserted with the movement

Restricted runtime SQL identity

DENY UPDATE / DELETE + SQL triggers

EF guard rejects history edits

An audit failure rolls back the money. A malicious administrator needs stronger controls.

Secure API boundaries. Reproducible deployment.

JWT roles + ownership | consistent errors | one-command assessment startup

ACCESS

Customer: own wallet / source

FundingSystem: inbound credit

Auditor / Admin: audit trail

JWT signature + claims validated

API OPERATIONS

Problem Details + trace IDs

Strict integer money inputs

Paginated newest-first history

JSON logs + per-instance limits

DOCKER COMPOSE

SQL Server

-> migrations / login bootstrap

-> restricted, non-root API

Liveness + database readiness

docker compose up

Disposable demo credentials only; no claim of production identity or real NIP integration.

Correctness demonstrated against real SQL Server

Recorded 16 September 2026 | 23 unit + 15 integration tests passed; zero failed/skipped.

100 competing spends

10 succeed / 90 reject; balances and evidence reconcile

50 duplicates / 2 hosts

One money movement + 49 response replays

Limits + boundary time

Contention respects cap; midnight WAT resets allowance

Audit fault + SQL writes

Whole transfer rolls back; historical edits are blocked

Deployed Docker smoke

Balances 9,000,000 / 1,000,000 kobo; replay + 409 verified

Correctness evidence, not a performance benchmark or production-readiness certification.

Simple scope. Explicit trade-offs. A credible next step.

Deliver the mandatory guarantees first; extend without weakening the transaction boundary.

DELIVERED NOW

- One application; one financial transaction
- Durable idempotency + immutable evidence
- 38 passing automated tests
- Compose + Swagger + deployed smoke

BEFORE PRODUCTION

- Managed identity, secrets + runtime upgrade
- Real settlement / full accounting design
- Admin tamper evidence + disaster recovery
- Load, lost-commit and deadlock fault tests

No broker or outbox in this submission. Add an outbox when external side effects exist.

Financial integrity belongs inside the transaction boundary. | Questions